EXAM PAPERS PRACTICE

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AQA A Level Further Mathematics (7367)

Topic	**DA: Graphs**
Sub topic	**DA5: Complete, Bipartite,and Complement Graphs**

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic DA: Graphs

Sub topic DA1: Complete, Bipartite, and Complement Graphs

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Name : ____________________

Date : ____________________

Time : ____________________

Total Marks Available : _____

Total Marks Received : _____

Q1.

(a) Draw a bipartite graph to represent the following adjacency matrix.

	1	2	3	4	5
A	0	0	0	0	1
B	0	0	0	1	1
C	0	1	0	1	1
D	0	0	0	1	1
E	1	1	1	0	1

(2)

(b) If A, B, C, D and E represent five people and 1, 2, 3, 4 and 5 represent five tasks to which they are to be assigned, explain why a complete matching is impossible.

(2)

(Total 4 marks)

Q2.

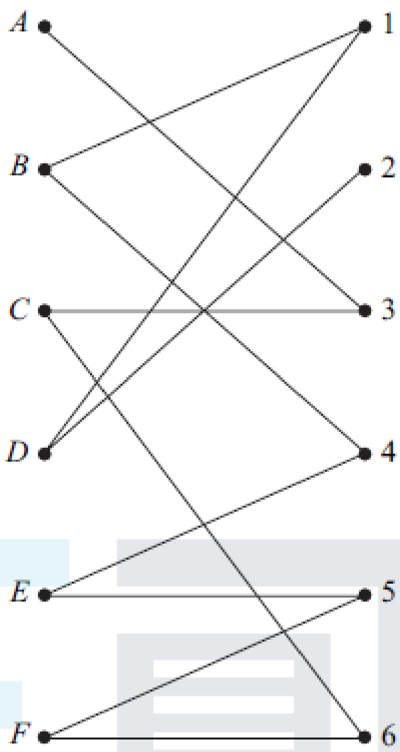
Draw a bipartite graph representing the following adjacency matrix.

	1	2	3	4	5	6
A	1	1	0	0	1	1
B	0	1	0	0	1	1
C	1	0	0	0	1	1
D	0	1	1	1	0	1
E	0	0	0	0	1	1
F	0	0	0	0	0	1

(Total 2 marks)

Q3.

Six people, A, B, C, D, E and F , are to be allocated to six tasks, 1, 2, 3, 4, 5 and 6.
The following bipartite graph shows the tasks that each of the people is able to undertake.



Represent this information in an adjacency matrix.

(Total 2 marks)

Q4.

The complete graph K_n ($n > 1$) has every one of its n vertices connected to each of the other vertices by a single edge.

(a) Draw the complete graph K_4.

(1)

(b) (i) Find the total number of edges for the graph K_8.

(ii) Give a reason why K_8 is not Eulerian.

(2)

(c) For the graph K_n, state in terms of n:

(i) the total number of edges;

(ii) the number of edges in a minimum spanning tree;

(iii) the condition for K_n to be Eulerian;

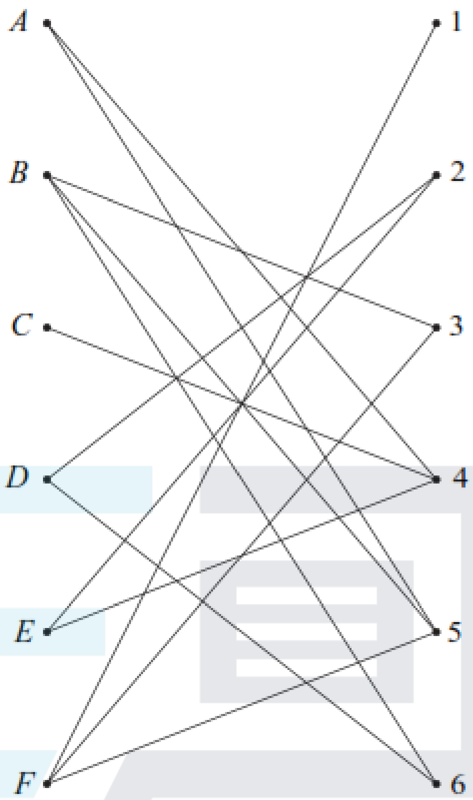
(iv) the condition for the number of edges of a Hamiltonian cycle to be equal to the number of edges of an Eulerian cycle.

(4)

(Total 7 marks)

Q5.

Six people, A, B, C, D, E and F, are to be allocated to six tasks, 1, 2, 3, 4, 5 and 6. The following bipartite graph shows the tasks that each of the people is able to undertake.



Represent this information in an adjacency matrix.

(Total 2 marks)

Q6.

(a) The complete graph K_n has every one of its n vertices connected to each of the other vertices by a single edge.

(i) Find the total number of edges in the graph K_5.

(1)

(ii) State the number of edges in a minimum spanning tree for the graph K_5.

(1)

(iii) State the number of edges in a Hamiltonian cycle for the graph K_5.

(1)

(b) A simple graph G has six vertices and nine edges, and G is Eulerian. Draw a sketch to show a possible graph G.

(2)

(Total 5 marks)

Q7.

(a) Draw the graph K_4.

(2)

(b) (i) Find the total number of edges in K_6.

(1)

(ii) State the number of edges in a minimum spanning tree of K_6.

(1)

(iii) Give a reason why K_6 is **not** Eulerian.

(1)

(c) For the graph K_n, state in terms of n:

(i) the total number of edges;

(1)

(ii) the number of edges in a minimum spanning tree;

(1)

(iii) the condition for K_n to be Eulerian.

(1)
(Total 8 marks)

Q8.

Draw a bipartite graph representing the following adjacency matrix.

	1	**2**	**3**	**4**	**5**
A	1	0	1	0	0
B	0	0	1	1	0
C	0	1	0	1	0
D	0	0	1	1	0
E	1	1	0	0	1

(Total 2 marks)

Q9.

A university mathematics lecturer is organising her six students into pairs to work on a mathematical modelling exercise. She would like to arrange it so that no students who have previously worked together are doing so on this exercise. So far

a has worked with b, d and e,
b has worked with a, d and f,
c has worked with d, e and f,
d has worked with a, b and c,
e has worked with a, c and f,
and f has worked with b, c and e.

(a) Draw a graph, G, to represent this information.

(3)

(b) Draw K_6, the complete graph on six vertices.

(1)

(c) Draw a graph on six vertices which consists of all those edges which are in K_6 but **not** in G. Say how this will help the lecturer.

(3)

(Total 7 marks)

Q10.

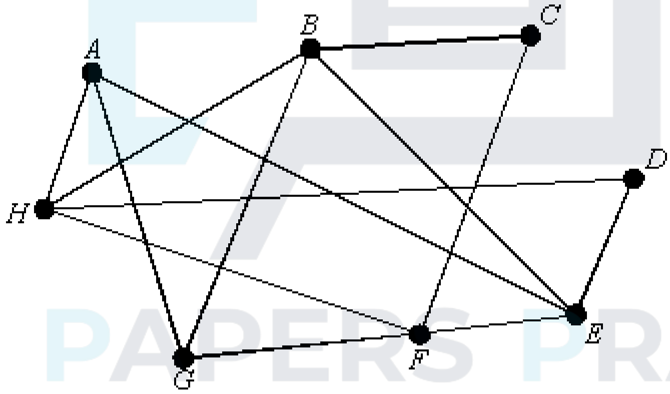
(a) Draw a sketch of the graph K_5.

(1)

(b) Draw a sketch of the graph $K_{3,3}$.

(1)

The graph **G** is shown:



(c) Draw a subgraph of **G** which is isomorphic either to K_5 or to $K_{3,3}$.

(2)

(Total 4 marks)